

Luis Martinez

Electrical Engineer

louisvizard.github.io/Portfolio · manilumart@gmail.com · (206) 475-7587

PROFESSIONAL EXPERIENCE

SPACE INFORMATION LABORATORIES

Electrical Engineer II

2025-Present

- Testing
 - Helped design electrical test fixtures and procedures for PCBA DVT
 - Designed system test equipment PCBAs to facilitate streamlined testing
 - Wrote test firmware for Xilinx SoC which assisted PCBA and module level DVT
 - Conducted GNSS receiver PVT accuracy and reacquisition characterization tests
- R&D Proposal Support
 - Assisted with proposal efforts by conducting trade studies and generating preliminary ICDs
- Circuit Analysis
 - Used worst-case circuit analysis (WCCA) to evaluate BMS circuit compliance to PMAP Rev B and RCC-319
 - Created FMECA reports for Autonomous Flight Termination Unit and BMS systems

RAYTHEON INTELLIGENCE AND SPACE

Electrical Engineer II

2024-2025

Electrical Engineer I

2022-2024

- PCB Design
 - Designed two boards based on customer requirements which were brought through CDR.
 - Used Hyperlynx SI/PI to run power and signal integrity of routed PCBs
 - Created PWB, PWA, parts list, and schematic drawings
- Circuit Analysis
 - Used worst-case circuit analysis techniques (WCCA) to evaluate the performance of various power electronics circuits
- Firmware Design
 - Implemented and debugged raw HDL for Intel SoCs in VHDL using SignalTap, Quartus Pro, and Qsys
 - Used Simulink and MATLAB for higher level system design
- Lab Experience
 - Conducted system level environmental tests with vibe/temperature chambers
 - Frequent work with signal generators, DC power supplies, spectrum analyzers, and oscilloscopes

BOSTON SCIENTIFIC

Electrical Engineering Intern

2021

- Designed a PCB to model behavior of the spine under certain stimuli
- Wrote C for a microcontroller on the PCB to define response to input data
- Used concepts of DSP, digital and analog filtering, and digital to analog conversion

PERSONAL PROJECTS

Guitar Headphone Amplifier

- Designed a 2-layer PCB which amplifies an electric guitar signal for use with a pair of in-ear headphones. Leverages the NE5532 op amp for active filtering and amplification.

GNSS and IMU PCBA with STM32 and RF Transceiver

- Designed a PCB centered around the NEO-M9N GNSS receiver, STM32F411 MCU, and BNO086 9DOF IMU. Includes an nRF24L01 2.4GHz RF transceiver with external PA/LNA circuitry

SKILLS

- | | |
|--------------------------------|-------------------------------|
| • Xpedition Enterprise | • FMECA |
| • Circuit Simulation (LTSpice) | • Python |
| • VHDL, Verilog, SystemVerilog | • Altium |
| • C/C++ | • Embedded Systems |
| • Matlab/Simulink | • VPX/VITA Standard Knowledge |

EDUCATION

CORNELL UNIVERSITY

Ithaca, NY

Bachelor of Science in Electrical and Computer Engineering

2018-2022

- Coursework centered around Embedded Systems, Digital Signal Processing, and FPGA development
- Electrical lead for Cornell Nexus which strived to create a robot that filters microplastics from beaches. Worked closely with motors and motor controllers, motor encoders, Raspberry Pi's, and batteries.